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- sample\_data
- Mall\_Customers (1).csv
- Mall\_Customers\_Clustered.csv
- heart\_disease\_uci.csv
- submission.csv
- test.csv
- test1.csv
- train.csv
- train1.csv

```
wcss.append(kmeans.inertia_)

plt.figure(figsize=(8,5))
plt.plot(range(1,11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()

# Train KMeans with 5 clusters
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
y_kmeans = kmeans.fit_predict(X)

# Visualize clusters
plt.figure(figsize=(8,6))
plt.scatter(X.iloc[:,0], X.iloc[:,1], c=y_kmeans)
plt.scatter(kmeans.cluster_centers_[0,0],
            kmeans.cluster_centers_[0,1],
            s=200,
            marker='X')

plt.title('Customer Segmentation')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.show()

# Add cluster labels
df['Cluster'] = y_kmeans

# Save output
df.to_csv('Mall_Customers_Clustered.csv', index=False)

print("Clustering completed successfully!")
```

Choose Files Mall\_Customers.csv

**Mall\_Customers.csv**(text/csv) - 3981 bytes, last modified: 6/2/2026 - 100% done

Saving Mall\_Customers.csv to Mall\_Customers (1).csv

|   | CustomerID | Gender | Age | Annual Income (k\$) | Spending Score (1-100) |
|---|------------|--------|-----|---------------------|------------------------|
| 0 | 1          | Male   | 19  | 15                  | 39                     |
| 1 | 2          | Male   | 21  | 15                  | 81                     |
| 2 | 3          | Female | 20  | 16                  | 6                      |

